

Brannock Chemicals

The base case said \$53 million and everyone in the room had stopped looking at it, which is how Ines Rautio ended up rebuilding the model on a Sunday.

Brannock Chemicals was evaluating a specialty resin plant on the Gulf Coast: \$310 million of capital, ten years of production, 92 million kilograms a year at a price of \$2.75 and a variable cost of \$1.55. Fixed cash costs of \$42 million, five-year MACRS on the whole \$310 million, a 25 percent tax rate, and Brannock's 10 percent cost of capital. Those numbers produced an NPV of \$65.1 million, and the capital committee had approved the project on them in September.

Rautio, who had been asked to prepare the funding paper for the board, had noticed that the model had a single column. Price was \$2.75 in every year and every scenario. Volume was 92 million kilograms the same way. The sensitivity analysis in the appendix moved one variable at a time by ten percent and reported that the project remained positive in all cases.

What bothered Rautio was that Brannock's own history did not look like that. She pulled fifteen years of realized price and volume for the two adjacent resin grades Brannock already made. Log price had a standard deviation of about 16 percent a year. Log volume about 22 percent. Neither series looked like a constant.

More to the point, the two series moved together, and Rautio could not decide whether they were supposed to. In two of the fifteen years, price and volume had both collapsed at once — 2009 and 2020, when the customers stopped building things. In three other years price had spiked while volume fell, which was what happened when a competitor's cracker went down and the market cleared on price. The first pattern is demand; the second is supply. Which one dominated the next decade was a question about the world, not about the spreadsheet, and it was going to change the answer.

Rautio asked the AI to build a Monte Carlo over the model. It produced a hundred thousand draws in under a minute, with lognormal price and lognormal volume calibrated to the standard deviations she gave it, and reported a mean NPV of \$53 million and a 46 percent chance of a loss.

She noticed that it had drawn price and volume independently. She had not told it not to.

Capital investment	\$310m
Project life	10 years
Volume	92m kg/yr
Price	\$2.75/kg
Variable cost	\$1.55/kg
Fixed cash costs	\$42m/yr
Depreciation	5-year MACRS
Tax rate	25%
Cost of capital	10%
Base case NPV	\$65.1m

Five-year MACRS, by year: 20.00, 32.00, 19.20, 11.52, 11.52, 5.76 percent. The plant is written off inside six years and the last four years of production carry no depreciation at all.

Historical volatility, from fifteen years of adjacent grades: log price 16 percent per year, log volume 22 percent per year.

ASSIGNMENT

- 1 Reproduce the deterministic base case. Confirm \$65.1 million. Report the after-tax operating cash flow year by year and say why it falls in year 7.
- 2 Build a simulation with lognormal price and volume at the volatilities given, drawn independently, with the means matched to the base case. Report the mean NPV, the standard deviation, the fifth percentile, the probability of a loss, and the probability that NPV is below negative \$150 million.
- 3 Rerun with a correlation of +0.65 between the price and volume shocks, and again with -0.65. Report the same five statistics each time.
- 4 The probability of a loss moves far less across the three correlations than the other statistics do. Explain why, and say which statistic you would put in front of the board.
- 5 At a correlation of +0.65 the mean simulated NPV is above the deterministic base case, even though every input distribution has its mean at the base case value. Where does the extra value come from, and is it real?